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Cytology & Cell Biology Module

Lecture 1

Membranous Organelles

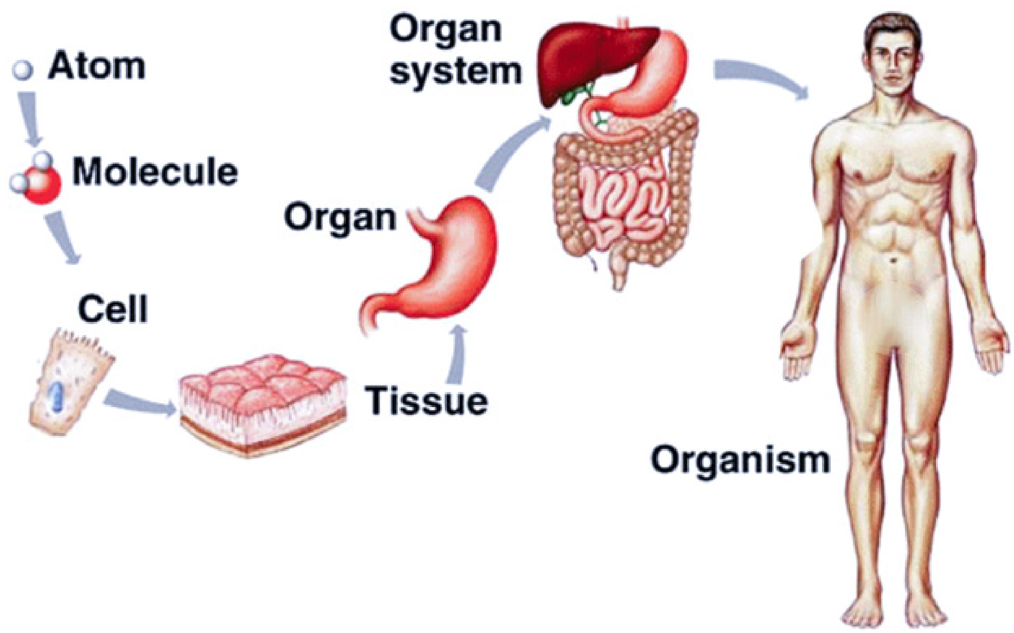
Membranous Organelles

HISTOLOGY

Study **microscopic structure** of cells, tissues and organs.

ORGANIZATION OF THE HUMAN BODY

1. **Cell:** smallest structural & functional unit in the body.
2. **Tissue:** similar cells having same characters and perform a particular function :
   - **Epithelial.**
   - **Connective.**
   - **Muscular.**
   - **Nervous.**
3. **Organ:** composed of the four basic tissues in various proportion to perform special function.
4. **System:** organs together perform complex function.



Figure(1): Levels Of Organization Of a Human Body.

The Cell

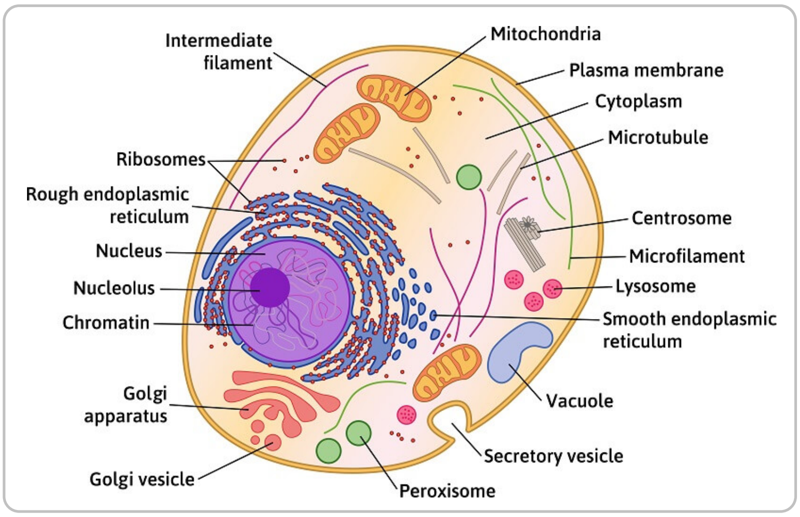
Cell membrane & protoplasm (**cytoplasm & nucleus**)

The cytoplasm contains:

1. **Cytosol:** a semifluid matrix.
2. **Organelles:** living , essential for the life of the cell & have function
3. **Cytoskeleton:** tubules and filaments (the shape of the cells, movement, intracellular pathways).
4. **Inclusions:** metabolic products, storage forms of nutrients, not essential for the cell life.

The Nucleus is formed of:

1. **Nuclear envelope.**
2. **Nucleolus.**
3. **Chromatin**
4. **Nuclear matrix**



Figure(2): Animal Cell Structre.

General scheme

اكتـــب ورايـــا

1 Cell membrane:

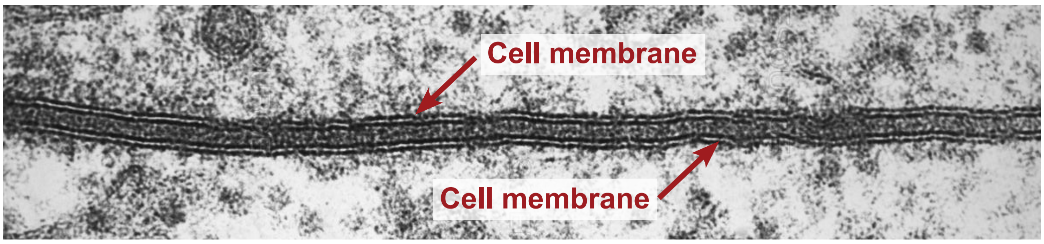
(plasma membrane or plasmalemma)

- **Definition:** عيب
- **Light microscope (LM):** Difficult to be seen (**7.5 : 10 nm**)
  - seen by **special stain** (**Silver Ag or PAS** فهم ??????)
- **Electron microscope (EM):**
  1. **Low magnification:** a dense thin line (7.5 - 10 nm) in thickness.
  2. **High magnification:** two electron dense lines separated by an electron-lucent intermediate zone

Trilaminar appearance (unit membrane).

NB. Fuzzy material on the outer surface only = **cell coat or glycocalyx**

Figure(3): Electron microscope (EM).



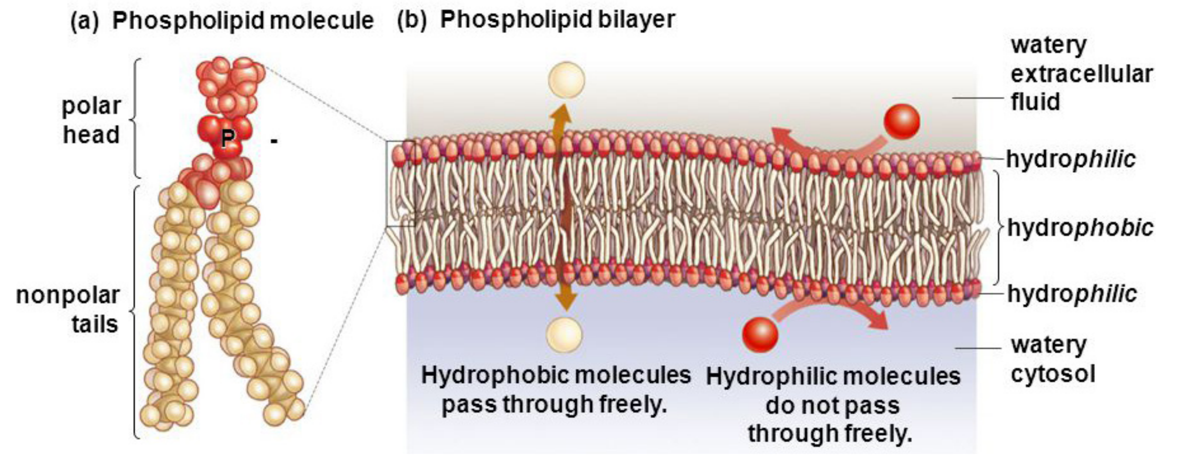
Figure(4): Cell membrane under Electron microscope (EM).

Molecular structure:

Lipid	Protein	Carbohydrates

1 Lipid:

A1 Phospholipids.
A2 Cholesterol.



Figure(5): Phospholipids Bilayer.

A1 Phospholipid bilayer:

the backbone of the plasma-membrane composed of 2 layers

1 Polar hydrophilic phosphate heads:

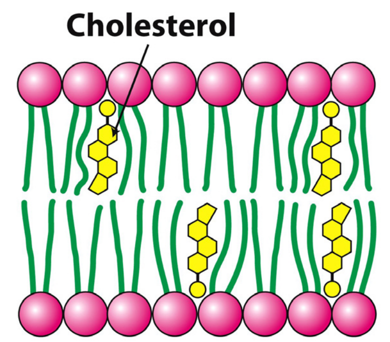
- Charged directed toward aqueous solution
- Facing outer & inner cytoplasmic surface

2 Non-polar hydrophobic tails of long fatty acid chains:

- Non charged directed inwards away from aqueous solution
- Facing each other

A2 Cholesterol molecules:

- have **1 to 1 ratio** with phospholipids.
- fill the gaps between the fatty acid tails and prevent their close packing.



Figure(6): Cholesterol molecules.

2 Proteins:

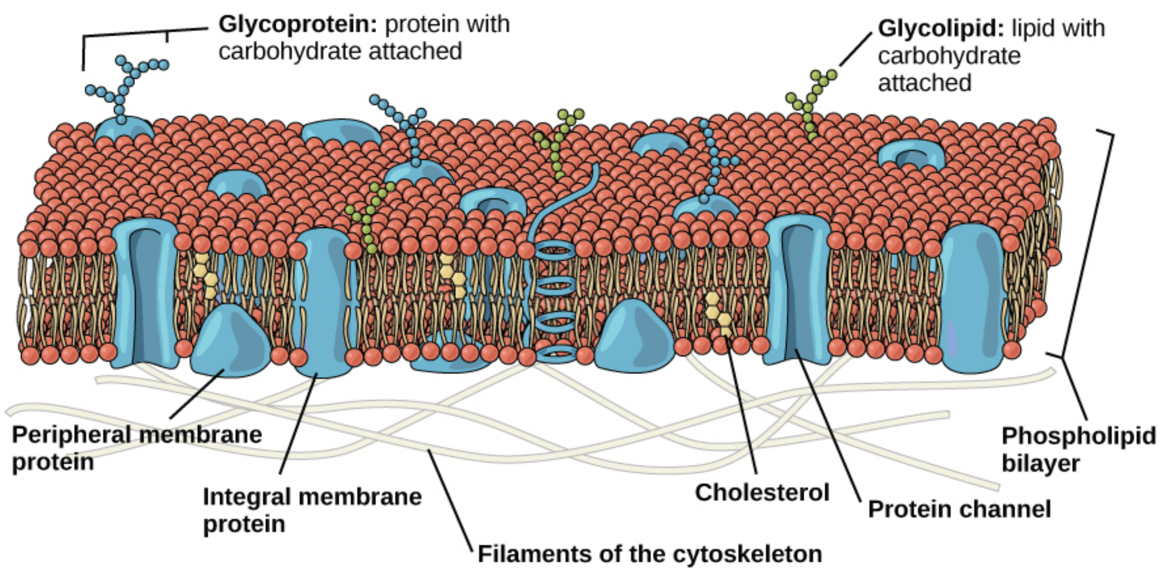
1 Integral (intrinsic) proteins:

- Embedded in lipid bilayer
- Some extend from side to side **(transmembrane protein)**

2 Peripheral Proteins:

- Associated with both surfaces.
- Loosely attached

3 Cell coat (Glycocalyx)



Figure(7): Cell coat (Glycocalyx).

Site : at external (outer) surface.
Structure:

- **Glycoprotein:** Oligosaccharide linked to protein
- **Glycolipids:** Oligosaccharide linked to phospholipid

NB. Acts as receptor for drugs ,hormones ,bacteria & viruses

Function: important in

- **Cell immunity**
- **cell protection**
- **cell recognition**
- **intercellular adhesions**

Functions of cell membrane:

1. **Exchange of material**
2. **Functions of cell coat**
3. **Modification of cell membrane**

   Microvilli: increase surface area for absorption
   Cilia: move particles in 1 direction
   Flagella: form tail of sperms
   Stereocilia: long microvilli
4. **Conduction of excitation waves**
   In nerve cells & muscle

1 exchange of materials

1 Passive diffusion :

- No energy is needed during this process.
- Depends on their concentration gradients. (**down**)
- **Fat soluble** & **small** substances
- **ex.** water , ions & dissolved gases

2 facillitated diffusion

- No energy is needed during this process.
- depends on their concentration gradients. (**down**)
- **Fat insoluble** & **small** substances
- ex. Glucose & amino acids

3 Active transport:

- Energy is required
- against their concentration gradient from low concentration to higher one .
- The process is carried out by carrier Protein
- e.g. Sodium-Potassium pump

4 Bulk transport

Definition: Taking up or releasing large substances.

1. **Endocytosis** : Bulk uptake of substances across the plasma membrane.
   - **Phagocytosis** (cell eating): Macrophages & Neutrophils can engulf:

     - Bacteria
     - Damaged cells
     - Protozoa
     - Unneeded extracellular constituent

Mechanism: when a bacterium bound to the surface of a macrophage - cytoplasmic processes extend and fuse-enclosing the bacterium in an intracellular **phagosome**.

- **Pinocytosis** (cell drinking): small membrane invaginations enclose extracellular fluid forming **pinocyticvesicles**.

- **Receptor mediated endocytosis:** Receptors for specific molecules (protein hormones & gross factors)are integral protein at the cell membrane.

ارسم معايا (فيديو)

2. Exocytosis: ???

Mechanism: A membranous cytoplasmic vesicles fuses with the cell membrane to release its content into the extracellular space .

NB. The vesicular membrane will be added to the plasma membrane

ارسم معايا (فيديو)

Choose the correct answer:

1 One of the following is not a character of cell organelles:

a. *They are metabolically active structures.*
b. *They perform distinctive functions.*
c. *They constitute intracellular pathways within the cells.*
d. *They are essential for the life of the cell.*

2 Cytoplasmic inclusions:

a. *Are storage forms of various nutrients.*
b. *Are formed of tubules and filaments.*
c. *Are essential for the life of the cell.*
d. *Maintain the shape of the cell*

3 As regard the cell membrane one of the following is not correct:

a. *It controls movement of substance in and out of the cell.*
b. *It is visible with light microscope.*
c. *It maintains the structural integrity of the cell.*
d. *It can establish transport system for specific molecule*

4 The characteristic trilaminar appearance of cell membrane is evident with:

a. *Light microscope.*
b. *Low magnification of E/M.*
c. *High magnification of E/M.*
d. *Fluid-mosaic model*

5 As regards phospholipids all are true EXCEPT:

a. *Formed of double layers.*
b. *They are more than cholesterol molecules.*
c. *They are arranged with their hydrophilic head outwards and hydrophobic tails inwards.*
d. *It is a backbone of the cell Membrane*

6 The phospholipid bilayer of cell membrane consists of:

a. *Hydrophilic heads outward and hydrophobic tails inwards.*
b. *Hydrophobic heads outward and hydrophilic tails inwards.*
c. *Hydrophilic tails outward and hydrophobic heads inwards.*
d. *Hydrophobic tails outward and hydrophilic head*

Put (true) or (false):

1. Cell membrane is visible with light microscope.
2. Cholesterol molecules of cell membrane are present in lipid bilayer.
3. Peripheral proteins of cell membrane are loosely attached to the membrane surfaces.
4. In active transport, molecules are transported against their concentration gradient.

Organelles		
	MEMBRANOUS	**NON-MEMBRANOUS**
Character	Covered by membrane Contain enzymes	Not covered by membrane Don't Contain enzymes
Examples	1. **Plasma membrane** 2. **Mitochondria** 3. **Endoplasmic reticulum** • Rough • Smooth 4. **Golgi apparatus** 5. **Lysosomes** 6. **Peroxisomes**	1. **Ribosomes** 2. **Cytoskeleton:** • Microtubules • Centrioles • Cilia • Flagella • Filaments (thin, intermediate) 3. **Proteasomes**

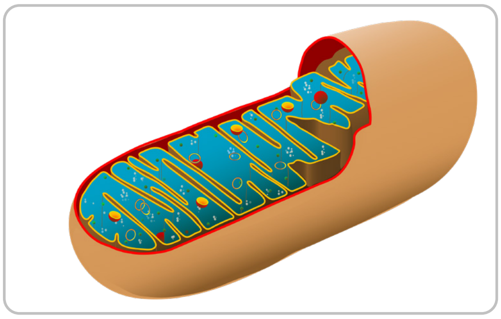
2 Mitochondria:

- **Definition :** Membranous organelles, containing enzymes responsible for aerobic respiration and energy production . They are the power-house of the cell **.**

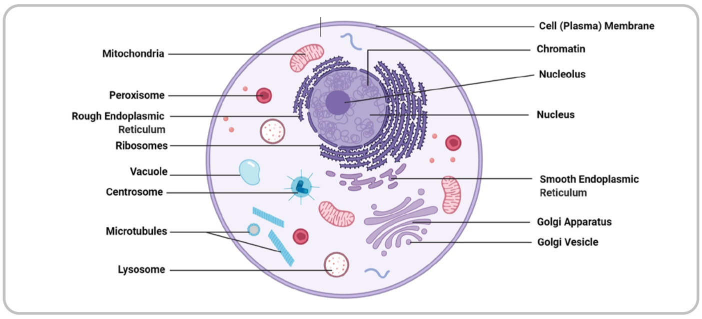
- **Sites :** at sites of most activity , where energy is needed.

- **Number :** - More numerous in **highly active** cell .
  - Varies according to the metabolic activity of the cell, e.g. a **liver cell** has about 1000 mitochondria, while a **lymphocyte** has few .

NB. Mitochondria can divide by simple division as they have their genetic apparatus (DNA).



Figure(8): .Mitochondrion.



Figure(9): Cross Section in animal cell.

LM:

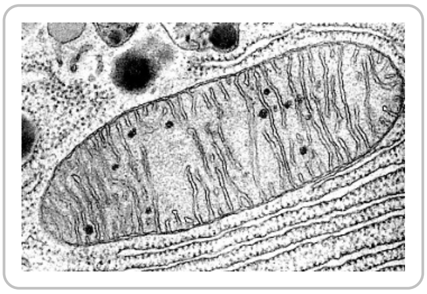
H&E: eosinophilic structures .

special stains:

- **Iron H:** dark blue **Janus green stain:** green

EM:

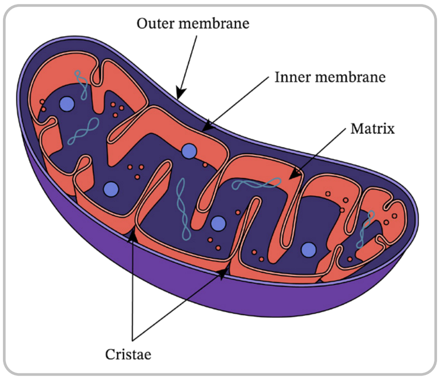
Rounded or oval membranous vesicles of **0.5-1 µm** in diameter and **10 µm in length**, surrounded with **two unit membranes** separated by an **inter-membranous space:**



Figure(10): EM Mitochondrion.

The outer membrane:

is smooth , and contains proteins called **porins** hence **permeable** .



Figure(11): inner & outer Membranes.

The inner membrane:

is less permeable(selective). it projects into the matrix forming shelf-like folds called **cristae.**

- Elementary particles are globular structures connected to the inner membrane by cylindrical stalks .
- the globular structures represent a protein complex with ATP synthetase activity (forms ATP in oxidative phosphorylation).

Mitochondrial matrix :

Composed of :
- Oxidative enzymes of citric acid cycle .
- A circular molecule of DNA .
- mRNA , tRNA and rRNA .
- Electron-dense granules rich in calcium ca++ that act as catalysts , regulating the activity of some mitochondrial enzymes.

Function :

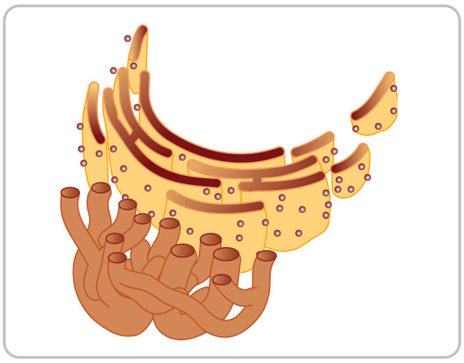
Cell respiration:

- Obtaining energy from metabolites present in cytoplasm by Krebs cycle .
- Most of this energy is stored as ATP molecules by **oxidative phosphorylation** and some is dissipated as heat to maintain body temperature.

3 Endoplasmic Reticulum:

- **Definition:** Communicating channels and sacs formed by a continuous membrane which encloses a space called a cisterna inside the cytoplasm .
- **Types :** According to presence or absence of electron dense particles (ribosomes) on the cytosolic side.

Rough	Smooth



Figure(12): Endoplasmic Reticulum

1 Rough endoplasmic reticulum (rER).

- **Site :** prominent in protein secreting cells e.g. plasma cell and fibroblast
- **LM :** localized or diffused **basophilia** in H&E stained section .
- **EM :** parallel flattened membranous **cisternae** studded with electron-dense particles, the **polyribosomes**, which are bound to specific receptors **(ribophorins)** on the membrane by their large subunits.
- **Function :**
  1. **Protein synthesis** by attached polyribosomes.
  2. **Segregation** of formed proteins .
  3. **Initial glycosylation** of some proteins by addition of some monosaccharides.
  4. **Packing** of formed proteins in membranous vesicles that bud off from the cisternae **(transfer vesicles)** to be delivered to Golgi apparatus .
  5. **Protection** of cytoplasm from hydrolytic enzymes formed inside its cisternae.
  6. **Intracellular** pathway for the formed substances

Figure(13): Rough Vs Smooth Endoplasmic Reticulum

2 Smooth endoplasmic reticulum (sER)

- **Site :** well-developed in fat & steroid hormones forming cells , e.g. liver cells & endocrine cells .
- **LM :** Not demonstrated , but when abundant causes cytoplasmic acidophilia .
- **EM :** Anastomosing tubular and variable shaped cisternae that lack bound polyribosomes. anastomose in an irregular manner and are continuous with rER.
- **Functions:**
  1. **Phospholipid** molecules **synthesis** that constitute cell membranes .
  2. **Steroid hormone synthesis** as cortisone & testosterone .
  3. **Utilization of glucose** originating from glycogen in liver cells .
  4. **Detoxification of drugs** , hormones & alcohol in liver cells .
  5. **Calcium ca^{++} ions release** (pump) in muscle contraction
  6. **Acts** as an **intracellular** pathway.

4 Golgi Apparatus :

- **Definition:** Golgi bodies are named after the Italian biologist Camillo Golgi . they are membranous organelles , concerned with secretion .
- **Site:** well developed in secretory cells.
- **LM:**

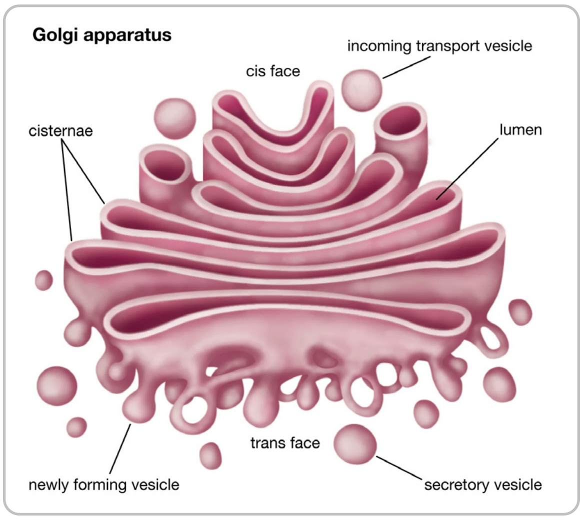
H&E: doesn't appear. in protein synthesizing cells (**plasma cells**) characterized by deeply basophilic cytoplasm , it can be seen as a **pale unstained area** near the nucleus called **"negative Golgi image"**.

With **silver stain (Ag)**= brown granules & fibrils , either:

- Apical in secretory cells .
- Perinuclear : in nerve cells .

- **EM:**

3-10 parallel flat curved membranous saccules arranged one above the other
Each saccule has a narrow lumen with expanded ends . It is filled with low electron dense material.



Figure(14): golgi Apparatus.

Saccules are inter-connected with each other .

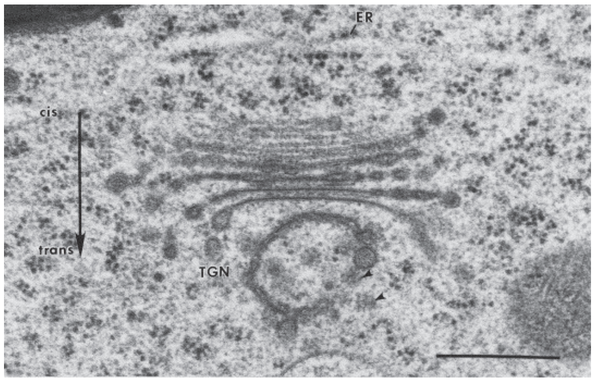
Each stack has 2 faces:

1. **Entry (Cis) face:**

   It receives **transfer vesicles** from rER carrying proteins , into the saccules to reach the trans face.

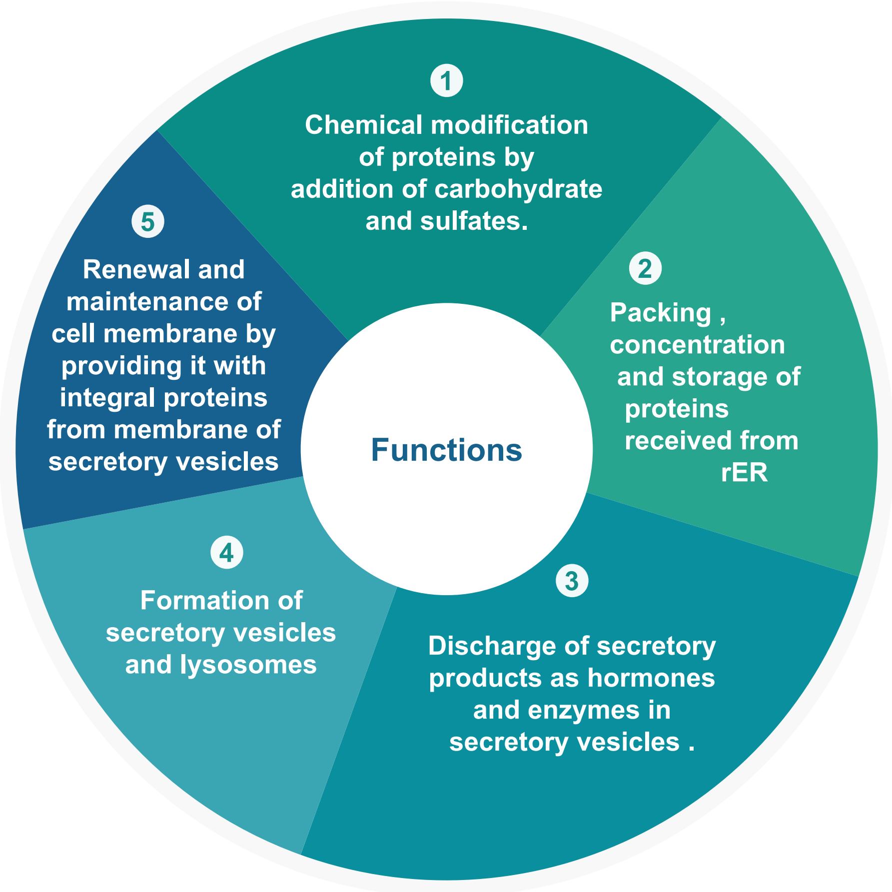
2. **Exit (Trans) face:**

   the vesicles that bud off from this face are either :
   (1) secretory vesicles that discharge their contents to the out side (exocytosis) or **(2) lysosomes**, which membranous organelles retained in the cytoplasm



Figure(15): EM Golgi Apparatus.

Functions:



Figure(16): Functions Of Golgi Apparatus

5 Lysosomes:

- **Definition:** Membranous vesicles, contain hydrolytic enzymes (phospholipase , protease , acid phosphatase , nuclease , sulfatase) concerned with intracytoplasmic **digestion.**
- **Origin:** the lysosomal hydrolytic enzymes are proteins synthesized with rER then carried to golgi apparatus in the form of transfer vesicles to come out in the lysosomes .
- **Number:** Abundant in phagocytic cells (macrophages and neutrophils).
- **LM:** Demonstrated by **histochemical tests** that detect the enzymes inside them as acid phosphatase enzyme. in **macrophages and neutrophils** they are large and can be seen by **LM.**
- **EM:**

1 Primary(1ry) lysosomes:

Newly released from golgi apparatus not entered into digestive events. Spherical vesicles surrounded with a single unite membrane. Moderately electron dense **homogenous**.

2 Secondary (2ry) lysosomes :

- When primary lysosomes **fuses with any intra-cytoplasmic vesicles**, it becomes a secondary lysosome .
- They appear **heterogeneous** in EM because of the digested materials.
- Differ according to the type of vesicle that 1ry lysosome fuses with.

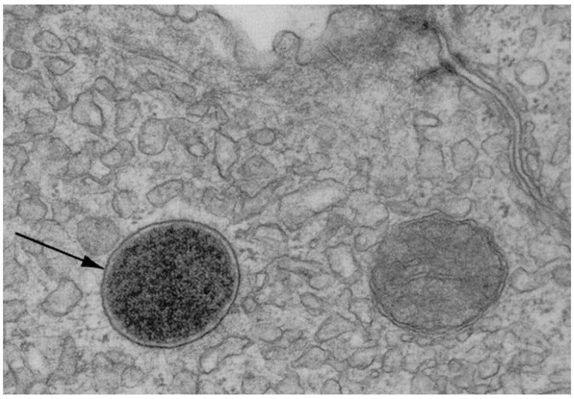
Type of secondary lysosomes :

1. **Heterolysosomes:** a 1ry lysosome fuses with **a phagocytic vesicle**. it empties its hydrolytic enzymes into the vesicle to digest its content (bacteria or viruses)
2. **Multivesicular bodies:** a 1ry lysosome fuses with **a pinocytic vesicle.**
3. **Autolysosomes:** A 1ry lysosome fuses with an **autophagic vesicle** which is a vesicle containing old organelles as old mitochondria , an endogenous component of the cell .
4. **Residual bodies:** undigested material is retained within vesicles.

   after digestion of contents of a 2ry lysosome , digested nutrients diffuse to the cytoplasm through lysosomal membrane , while undigested material is retained within vesicles.

Fate of residual bodies:

1. Discharge their contents by exocytosis (cytosol).
2. Accumulate within the cells as lipofuscin granules (age pigments)in long lived cells (nerve cells and heart muscles).



Figure(16): Lysosomes Under Electron microscope

- **Functions of lysosomes:**
  1. **Digest nutrients** and foreign invaders , as bacteria and viruses.
  2. **Maintain cell health** by elimination of old or worn out organelles.
  3. **Postmortem autolysis** by digestion of the whole cell after death due to escape of the hydrolytic enzymes from the lysosomal membrane .

Japanese cell biologist Yoshinori Ohsumi

  4. **Fertilization** by helping the sperm to penetrate the ovum.

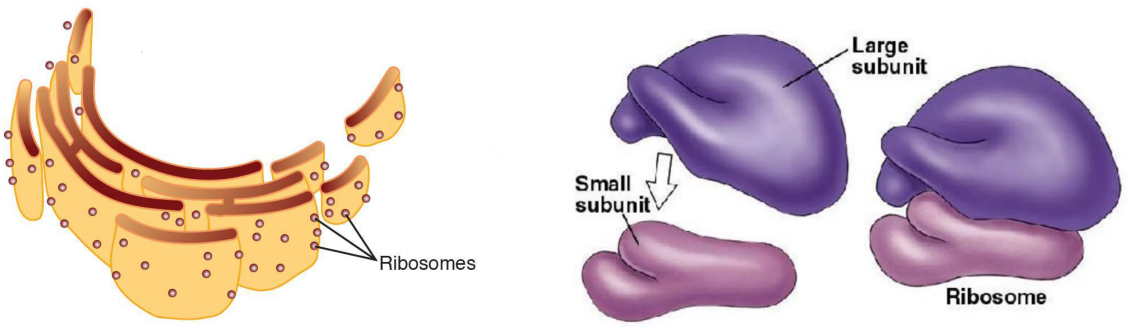
6 Peroxisomes (Microbodies):

- **Definition:** peroxisomes are membranous organelles that contain **enzymes of beta oxidation** of long chain fatty acids and **catalase** .
- **Origin:** - precursor vesicles bud off the **rER.**
  - Peroxisomal enzymes are synthesized in free polysomes.
- **LM:** Can't be demonstrated.
- **EM:** Small (**0.5µm in diameter**), spherical to ovoid , bound by a single membrane.
- **Functions:**
  1. **Beta oxidation of long chain fatty acids**, forming **acetyl coenzyme A (CoA)** and **hydrogen peroxide (H2O2)**.
     - Acetyl coenzyme A (CoA) Is used by the cell for metabolic needs.
     - Hydrogen peroxide detoxifies various noxious agents (e.g, ethanol) and kills microorganisms.
  2. **Catalase** reduce excess hydrogen peroxide to water and oxygen.
  3. **Energy produced** comes out as heat and not stored as ATP .

Lecture 2

Non Membranous Organelles

1 Ribosomes



Figure(1): Diagram Of Ribosome on rER & Ribosome structure

- **Definition:** Non-membranous electron dense particles (**20×30nm**).
- **Origin:** formed of strands of rRNA synthesized in nucleolus and proteins synthesized in cytoplasm.
  Proteins and rRNA associate in nucleus into small and large subunits that pass to the cytoplasm for protein synthesis .
- **Number:** Abundant in protein synthesizing cells.
- **LM:** when abundant they cause cytoplasmic **basophilia** due to RNA .

Cytoplasmic basophilia may be :

1. **Diffuse:** As in embryonic cells.
2. **Localized:** As in pancreatic cells (at the base of cell).
3. **Focal:** As in nerve cells (Nissl's granules).

- **EM:** Small electron dense granules , each is formed of 2 subunits , small &large . the 2 subunits unite together by binding to mRNA .
- **polysomes (polyribosomes):** are linked together by mRNA ,so appear as rosettes or spiral chains .

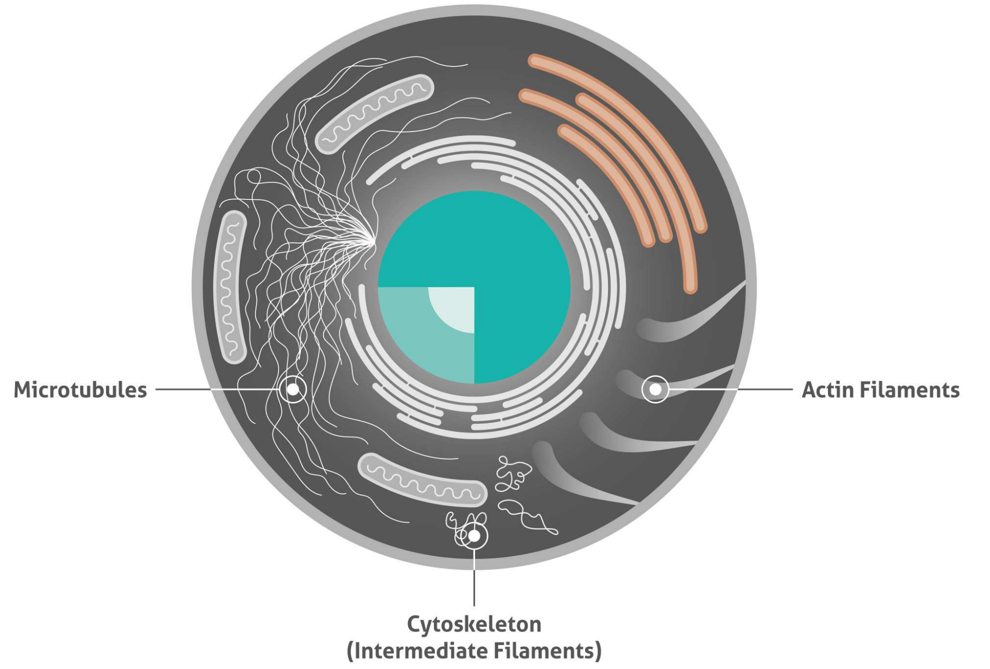
Two forms:

1. **Free :** Scattered singly & as polyribosomes.
2. **Attached :** Bind to the outer surface of rER by the large subunits at glycoprotein receptors (ribophorins).

- **Function:**

1. **Factories of protein** synthesis .
2. **Free ribosomes** form proteins used within the cytosol as glycolytic enzymes .
3. **Attached ribosomes** form proteins secreted by the cells as enzymes and hormones .

2 Cytoskeleton:



Figure(2): Diagram Showing Cytoskeleton

- **Definition:** Complex network of microtubules and filaments , together with some proteins to link them to each other and to the cell membrane forming a framework .

A Microtubules (MTs):

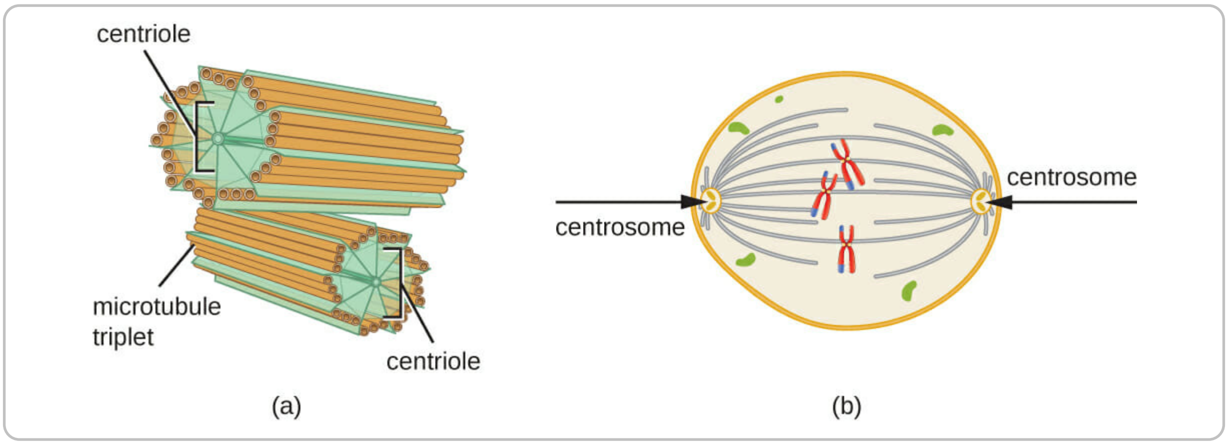
- **Definition :** Fine tubes of constant diameter (24nm) ,but unfixed length .
- **LM :** They are difficult to see by LM except by using immunofluorescent technique.
- **EM :** They appear as fine tubules formed of alpha and beta tubulin protein molecules that arrange into 13 protofilaments .their length varies according to polymerization of tubulin molecules.
  this is directed by **microtubular organizing center (MTOC)**.
- **Forms :**
  Dynamic: With continuous assembly and disassembly causing reshaping of cells.
  Stable: Form the walls of centrioles, cilia and flagella.
- **Functions :**
  1. Determination of the **shape** of the cell.
  2. **Intracellular transport** of macromolecules.

3. Formation of the **mitotic spindle** during cell division.
4. Formation of **centrioles, cilia** and **flagella**.

Structures formed by stable microtubules.

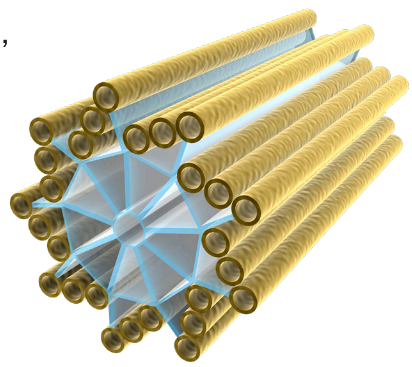
1 Centrioles:

- **Definition:** cylindrical structures composed of highly organized MT.
- **LM:** Demonstrated with iron hematoxylin stain as two dark bodies near the nucleus .

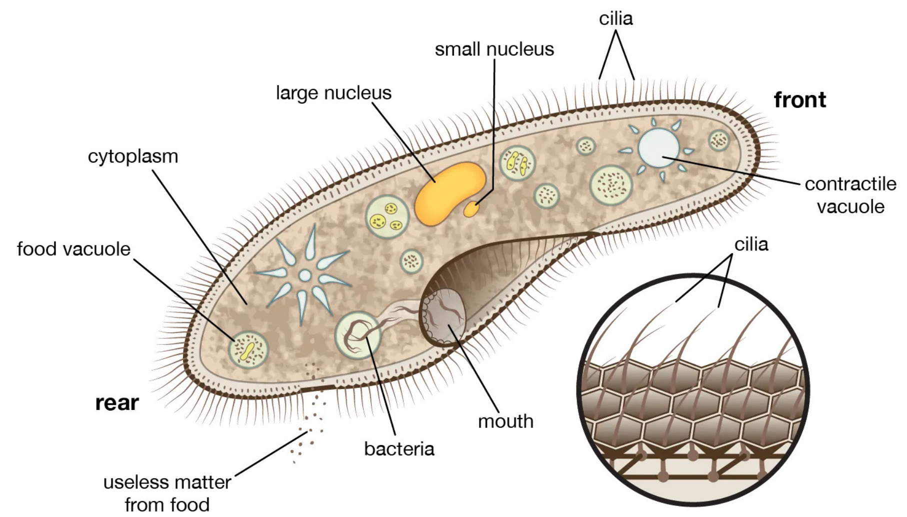


Figure(3): Diagram Showing Centrioles

- **EM :** 2 cylindrical structures , perpendicular to each other and surrounded by a matrix of tubulin (centrosome) in nondividing cells .
  - The wall of each cylindrical is formed of **9 bundles** of MT , each bundle is formed of 3 MTs **(triplets)** . consequently , the wall of the centriole is formed of 9×3 **=27 MTs** . MT (A) is complete and formed of 13 protofilaments . MTs (B&C) share protofilaments , thus each is formed of 10 protofilaments .
- **Functions:**
  1. It has a role during **cell division** in the cell cycle, the centrosome duplicates itself .
  2. Share in the formation of **cilia and flagella** .



2 Cilia:



Figure(4): Diagram Showing cilia

- **Definition :** Cilia are motile processes with highly organized MT core and covered by cell membrane.
- **Origin :** Centrioles duplicate thousands of times to form basal bodies of cilia that migrate to the apical cytoplasm. A shaft grows up from each cilium .
- **LM:** Hair-like striations.
- **EM:** Formed of 3 parts : a

Basal body
Shaft
Rootlets

1 Basal body:

- A single centriole formed of **27 MT** in **9 triplets.**
- It is embedded in the cytoplasm.

2 Shaft (axoneme):

- From each triplet , A&B MTs, grow as doublets pushing the cell membrane in front of them.
- So, the shaft is formed of **9 doublets** as its periphery in addition to 2 singlets MTs in the center formed by polymerization = **20 MTs.**

3 Rootlets:

- they are formed by growth of the third MT C in each triplet of the basal body into the cytoplasm (I.e., 9 MTs). They fix the basal body and shaft to the cytoplasm.

- **Functions:** Move secretions or particles over the tissue surface in one direction e.g. in respiratory and female genital systems.

3 Flagella:

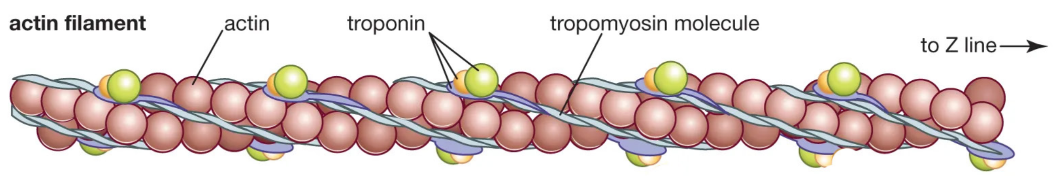
- They are motile projections from the cell, designed to move the cell itself.
- They have exactly the same structure as the axoneme of the cilium (nine peripheral doublets and 2 central singlets), but extremely longer. In human , a flagellum forms the tail of the sperm , which helps its movements.

B Filaments:

- **Definition:** minute threads that act as a part of the cytoskeleton.
- **Types:** According to size, two forms are present :

1 Microfilaments (actin filaments):

- **Definition :** Fine strands (**5-7nm**) of 2 chains of globular G actin, coiled around each other to form filamentous F actin.



Figure(5): Diagram Showing Actin Monomer

- **Sites & functions:**
  1. **In most cells:**
     - Formation of a band or network beneath the cell membrane involved in cell shape changes as endocytosis, exocytosis and amoeboid movements.
     - Movement of organelles, vesicles and granules (**cytoplasmic streaming**)
  2. **In dividing cells:** interact with myosin for cleavage of mitotic cells.
  3. **in skeletal muscles:** interact with myosin filaments for contraction.
  4. **in cell provided with microvilli:** formation of microvilli core to keep their shape and to change their length by shortening and elongation.

2 Intermediate filaments:

- **Definition:** A family of filaments (**8-10nm**) in diameter.

- **Types & functions:** They serve for support.
  1. **Cytokeratin:** found in epithelial cells and form hairs and nails.
  2. **Vimentin:** in connective tissue and vascular smooth muscle.
  3. **Desmin:** in muscle cells to join myofibrils together.
  4. **Neurofilaments:** in neurons.
  5. **Glial filaments:** in glial cells.
  6. **Lamins:** in nuclear envelope

Thick filaments: myosin filaments in muscle 10 nm in diameter.

Cytoplasmic inclusions

- **Definition:** Non-living, temporary, nonessential, metabolically inert cytoplasmic contents that result from cell activity.

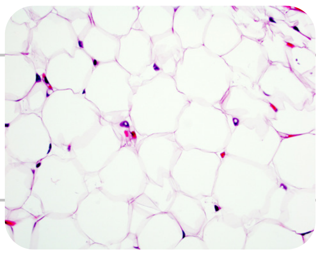
A Stored food

1 Carbohydrates

- **Site:** stored as glycogen granules in liver and muscle cells.
- **LM:**
  - **H & E:** Vacuoles.
  - **Best's carmine stain:** Red granules.
  - **PAS:** Magenta red granules.
- **EM:**
  - Single granules (**β granules**) or rosette-shaped aggregations (**α granules**)
  - Mostly concentrated in areas of cytoplasm rich in sER.

2 Fats

- **Site:** in fat cells as large globules or on liver cells as small droplets.
- **LM:**
  - **H & E:** fat appears vacuolated as it dissolves during preparation signet ring appearance.
  - **Sudan III** Orange and Sudan black stain: black globules



B Pigments

- **Definition:** Colored particles either produced by the cells or taken from out side. Pigments are materials that possess color of their own nature.
- **Types:**

1 Endogenous:

- **Hemoglobin (Hb):** in RBCs, carries gases (O_2 & CO_2).
- **Melanin:** in skin, to give its color & protects from ultraviolet rays.
- **Lipofuscin:** in cardiac muscle & nerve cells, they are waste products which accumulate with age.

2 Exogenous:

- **Carbon & Dust** particles: in dust cells of lung.
- **Carotene** pigments: in carrots.
- **Tattoo marks:** Certain dyes are injected under the skin & taken by phagocytic cells.

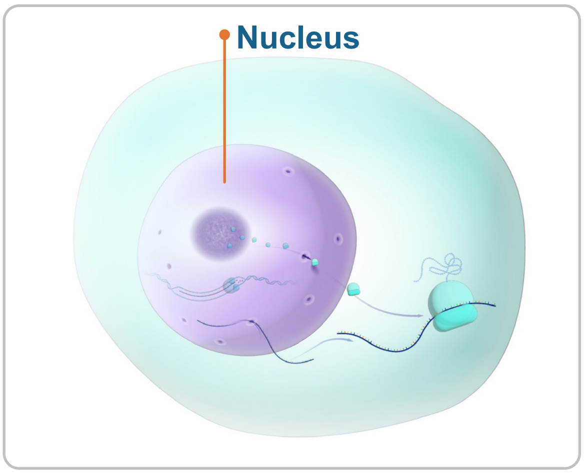
Lecture 3

Nucleus

Nucleus

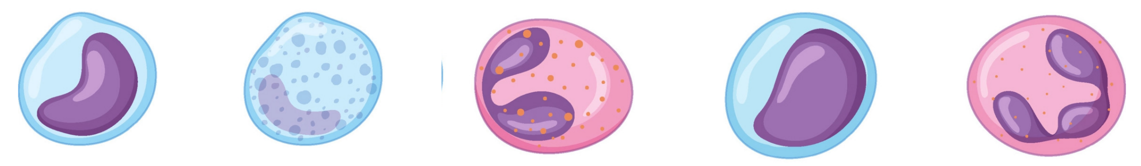
- **Definition:**
  The largest and most distinct component of all cells, except RBCs and platelets (not true cells).
- **Number:**
  Usually one nucleus is present in each cell (mononucleated).

  Some cells have two nuclei (**binucleated**) as liver cells.
  Some have more than two nuclei (**multinucleated**) as osteoclasts & skeletal muscles.



Figure(1): Diagram Showing Nucleus

- **Position:**
  usually Central, may be eccentric, basal or peripheral. The nucleus is always present in **the wider part** of the cell.
- **Shape:**
  Rounded, flattened, oval, kidney-shaped, segmented, lobulated, or bilobed (horse shoe).



Figure(2): Diagram Showing WBCs with different shapes of nucleus

- **Staining:**
  Basophilic dark blue with basic stain as hematoxylin or methylene blue.
- **Structure of the nucleus:**
  1. **Nuclear membrane (envelope).**
  2. **Chromatin material.**
  3. **Nucleolus.**
  4. **Nuclear sap**.

Structure of the nucleus

1 Nuclear Membrane (Nuclear Envelope):

- **LM:** blue (basophilic) line due to chromatin on its inner side & ribosomes on its outer side.
- **EM:** double walled membrane formed of two parallel unite membranes separated by a perinuclear space (30-50nm) and interrupted at intervals by nuclear pores.

1 Outer nuclear membrane:

It is rough and granular due to attached polyribosomes and is continuous with **rER cisternae**.

2 Inner nuclear membrane:

it is fibrillar due to attached chromatin threads (**peripheral chromatin**). It is associated with **nuclear lamina** formed mainly of **lamins** for stabilization of nuclear envelope.

3 Nuclear pore complexes:

Are circular opening at intervals, where inner and outer membrane fuse. **nucleoporin filaments** extend into the cytoplasm and nucleus.

- **Function:** Transport of proteins to the nucleus & export of RNA and ribosomal subunits to the cytoplasm through receptor proteins.

2 Chromatin:

- **Definition:** In non dividing nuclei chromatin is the chromosomal material in uncoiled state. it represents the genetic material, formed of nucleoproteins (DNA + Histones).
- **LM: Basophilic**

having two forms according to cell activity:

1. **Extended or active:**
   Appears pale basophilic (vesicular nucleus) as it represents extended (un-coiled) chromosomes and contain the active genes that direct protein synthesis in protein-forming cells.
2. **Condensed or inactive:**
   Appears dark basophilic (condensed nucleus) as it represents coiled chromo-somes and contain the inactive gene eg. Small lymphocytes.

- **EM:**
  1. **Euchromatin:** Appears as fine granules.
  2. **Heterochromatin:** Appears as coarse granules.

Site of heterochromatin:

1. **Peripheral chromatin:** attached to the inner surface of the nuclear membrane.
2. **Chromatin islands:** aggregated clumps scattered in the nuclear sap.
3. **Nucleolus-associated chromatin:** condensed around the nucleolus.

- **Functions:**
  1. Carries genetic information.
  2. Directs & control protein synthesis.
  3. Formation of RNA (mRNA,tRNA,rRNA).

3 Nucleolus

- **LM:**

Rounded, deeply basophilic mass rich in nucleic acid and surrounded with chromatin. Usually one or two in each nucleus.

- **EM:**

Spongy appearance, not limited by membrane. It has **dark and light areas**.

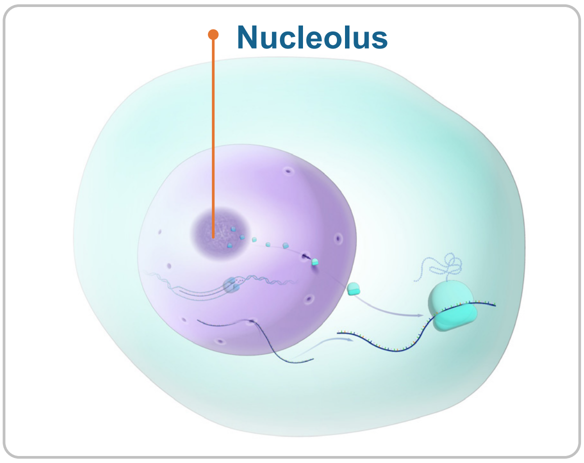
The dark areas are formed of:

1. **Pars amorpha (nucleolar organizer):** filaments of DNA encoding rRNA.
2. **Pars fibrosa (fibrillar component):** strands of newly formed rRNA.
3. **Pars granulosa (granular component):** granules of mature rRNA.

Both pars granulosa and fibrosa are called **"nucleolonema"**.

The light areas:
formed of the nucleolar sap.

- **Functions of the nucleolus:**
formation of **rRNA**



Figure(3): Diagram Showing Nucleolus

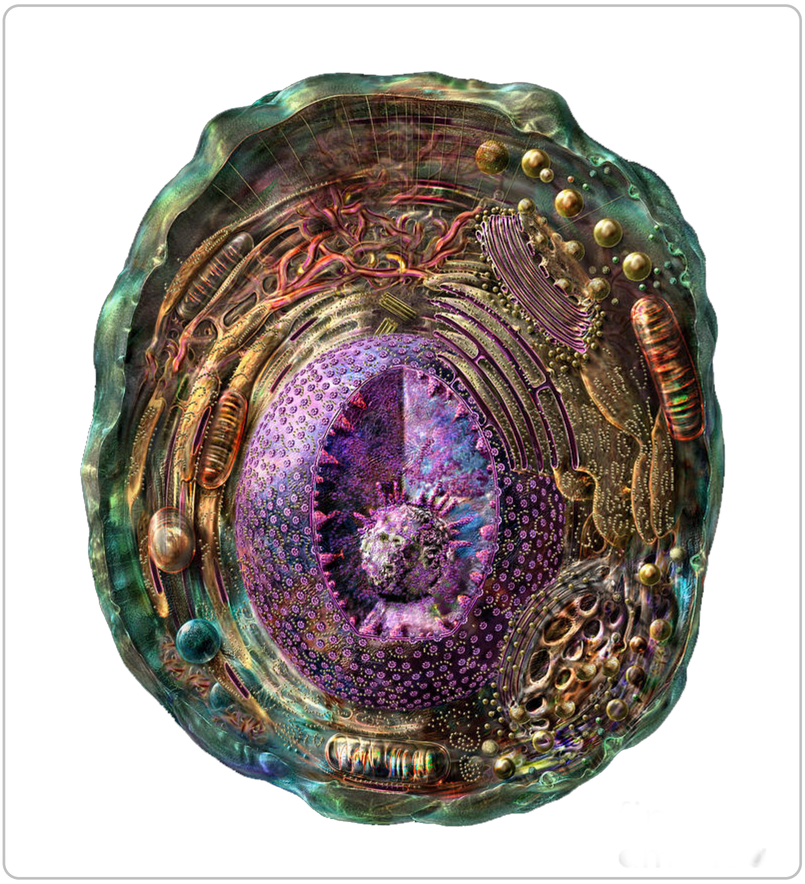
4 Nuclear sap

- **Definition:**
  a colloidal solution that fills the spaces between chromatin material and nucleolus.

- **Constituents:**
  formed of nucleoproteins, enzymes, sugars, calcium, potassium & phosphorous ions.

- **Function of nuclear sap:**
  provides a medium for transport of RNA through nuclear pores to cytoplasm for protein synthesis.

- **Function of the nucleus**
  1. **Carries all the genetic information** and hereditary factors.
  2. **Controls all the cell functions** including protein synthesis.
  3. **Responsible for** the formation of RNA.
  4. **Direct cell division.**



Figure(4): 3D Simulation Of Animal Cell